

//

1. What are the advantages of Polymorphism?

- More adaptable code
- Easier updates and extensions
- Cleaner and simpler code structure
- Runtime flexibility

2. How does Inheritance help make Polymorphism possible in Java?

- Inheritance allows one class (a subclass) to build upon another class (the superclass). This is important for polymorphism because when a subclass overrides a method from the superclass, Java can decide at runtime which version of the method to run, even if the object is being referred to by the superclass type.

3. What's the difference between Polymorphism and Inheritance in Java?

Concept	Polymorphism	Inheritance
What it is	A way for objects to behave in multiple ways	A way to pass down code and behavior from one class to another
Main goal	To allow the same interface to work with different data types or objects	To reduce duplication and define relationships between classes
Types	Method overloading (compile-time), method overriding (runtime)	Includes single, multilevel, and hierarchical inheritance
How they relate	Polymorphism depends on inheritance to override methods	Inheritance doesn't require polymorphism, but it enables it
Use case	Useful when different objects need to respond differently to the same method call	Useful when multiple classes share common features or actions